

From Obstruction to Design: Fibre Conditions for Monitoring and Indicator Choice

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Abstract

The obstruction calculus says, given a monitoring structure, why viability fails; governance needs the converse — what must be measured for viability to be possible. This paper, the D8 item of the generalization catalogue, develops the design reading on the programme’s working systems. The fibre condition — every observation fibre must contain states whose floor-servable instrument sets intersect — is proved necessary and, in the intersection form, necessary-and-sufficient for one-step admissibility of every singleton belief (verified over all 15 partitions of the four-state target region); the pairwise form of the rule is shown by an exact three-codex instance to be *necessary but not sufficient* off convexity — the one-dimensional Helly projection fails for non-convex instrument codices, with the intersection certificate surviving. Three design rules follow as exact identities: the delay rule (an indicator is acceptable only with period beating worst-case exit — viability at deadline T_{obs} is exactly the identity $z_0 \geq 1 + T_{\text{obs}}$, equivalently $T_{\text{obs}} \leq \sigma^*(z_0) = z_0 - 1$, evaluated exactly on all 48 grid cells; the boundary case $z_0 = 1 + T_{\text{obs}}$ is viable — the review lands exactly as the floor is reached — so the strict form $T_{\text{obs}} < \sigma^*(z_0)$ of the earlier editions is false on the grid; the rule rejects wherever exit beats the review period), the bias rule (a biased reading is acceptable only under correction: the canonical uncorrected law’s $\frac{21}{100}$ drift bound, the corrected law’s drift identically zero), and the aggregation rule (an aggregate index is acceptable only if its fibres do not merge policy-incompatible states: the audited belief is aggregate-nonviable, z_1 -only-viable). The coarsest admissible monitoring of the target region costs exactly two cells (7 of 15 partitions admissible, enumerated exhaustively), by exhaustive enumeration (8/8 check families).

Introduction and relation to the programme

The completed items diagnose; the source catalogue reverses the arrow, and this paper executes the reversal: obstruction as design. The fibre condition, the coarsest-admissible-monitoring problem, and the three indicator

rules (delay, bias, aggregation) are stated in the catalogue and here made exact and verified on the audited systems. The successor’s information lattice supplies the semantics of structures; the monitoring companion changes nothing in it.

The fibre condition

For a state x let $\mathcal{R}(x)$ be its floor-servable instrument set. For an observation structure with fibres F :

Proposition 1 (Intersection form: necessary and sufficient). A structure — a partition of the target region — is one-step admissible iff every fibre F of the partition satisfies $\bigcap_{x \in F} \mathcal{R}(x) \neq \emptyset$; equivalently, iff every belief the structure can present — each fibre, viewed as the information state of its reading — is one-step admissible. In particular a state x whose fibre is the singleton $\{x\}$ is one-step admissible exactly when $\mathcal{R}(x) \neq \emptyset$. The pairwise form ($\mathcal{R}(x) \cap \mathcal{R}(y) \neq \emptyset$ for all $x, y \in F$) is necessary but not sufficient in general.

Proof. Necessity and sufficiency of the intersection form are the read-then-act requirement itself: the cell’s one instrument must serve all of its states, and a belief presented by the structure is admissible exactly when its fibre is servable. For the pairwise claim: on non-convex codices three pairwise-intersecting sets can have empty total intersection; the exact instance below realizes this, so pairwise compatibility cannot imply the existence of a common instrument. (In one dimension with convex sets, Helly’s theorem would protect the pairwise form — the instance shows exactly where that protection ends.) $\square$

On the audited target — four states of the two-patch system (Abaee 2026a, Section 8; instruments $u \in \{1, 2\}$ protecting one patch, damage 2 on the other) — the floor-servable instrument sets are $\mathcal{R}(1, 2) = \{1\}$, $\mathcal{R}(2, 1) = \{2\}$, $\mathcal{R}(2, 2) = \{1, 2\}$, $\mathcal{R}(3, 1) = \{2\}$.

Remark 1 (Scope of the within-target reading). The structures audited here are partitions of the target region itself, so every fibre lies within the target by construction. For a general observation map whose fibres cross the target boundary, the intersection test must range over the *whole*

fibre — a state outside the target bounds the admissible action exactly as one inside — and the full fibre criterion (Abaee 2026a, Proposition 7) governs. The within-target form is a design-space restriction, not a claim about crossing fibres.

Proposition 2 (Dynamic caveat). Singleton fibre-admissibility is necessary only: a structure admissible for every singleton belief can still obstruct a multi-state belief dynamically. On the audited target the exhaustive search records whether such an instance exists; either outcome is an exact finding.

Proof. Successor cells re-partition, so one-step admissibility of singletons does not compose into multi-step viability; the dynamic statement is governed by the recursion, not the static condition. $\square$

Verified record (9/9 exact check families)

E1. Over all 15 partitions of the target $\{(1, 2), (2, 1), (2, 2), (3, 1)\}$: the intersection form is equivalent to singleton admissibility, instance by instance. The non-convex counter- instance: $\mathcal{R}_1 = \{\frac{2}{10}, \frac{6}{10}\}$, $\mathcal{R}_2 = \{\frac{2}{10}, \frac{65}{100}\}$, $\mathcal{R}_3 = \{\frac{6}{10}, \frac{65}{100}\}$ — pairwise intersections nonempty, total intersection empty.

E2 (an exact absence). The dynamic caveat’s exhaustive search over 15 partitions $\times$ target pairs found *no* instance on this target: the singleton rule happens to be sufficient here; the absence is recorded as the finding (necessity-only remains the general claim).

E3. The coarsest admissible monitoring costs exactly 2 cells (7 of 15 partitions admissible; costs 2, 2, 3, 3, 3, 3, 4). Two 2-cell partitions are minimal: the isolating partition $\{\{(1, 2)\}, \{(2, 1), (2, 2), (3, 1)\}\}$ — the demanding state (1, 2) alone, the rest merged — and the crossing partition $\{\{(1, 2), (2, 2)\}, \{(2, 1), (3, 1)\}\}$, which pairs (1, 2) with the flexible state (2, 2) instead of isolating it. The remaining five admissible partitions are refinements of these two (the admissible set is the upper set they generate); enumeration certifies minimality and lists all seven. A scoping correction (check E9): the maximal common-action sets of the target, $\{(1, 2), (2, 2)\}$ and $\{(2, 1), (2, 2), (3, 1)\}$, *overlap* in (2, 2), so the coarsest adequate monitoring exists (by the upper-set property), is not unique (two minimizers, above), and is *not* “the” partition into maximal common-action sets — no such partition exists on this target.

E4 (delay rule). On the 48-cell grid with the declared hold class: viability at deadline $T_{\text{obs}} \in \{1, 2, 3\}$ is exactly the identity $z_0 \geq 1 + T_{\text{obs}}$, i.e. $T_{\text{obs}} \leq \sigma^*(z_0) = z_0 - 1$ — the review must land no later than worst-case exit; the boundary cell $z_0 = 1 + T_{\text{obs}}$ is viable (verified, check E9), so the strict inequality of the earlier editions fails there and is corrected in this edition.

E5 (bias rule). The trap’s uncorrected reading drives the canonical certainty-equivalence law to drift $\geq \frac{21}{100}$ (exit); correction restores drift identically zero — biased

indicators are acceptable only under correction. (The quantifier over the whole uncorrected class is the successor architecture’s layer-five reading, cited not re-claimed.)

E6 (aggregation rule). The audited belief $\{(1, 2), (2, 1)\}$ is nonviable under the aggregate index and viable under the z_1 -only index: aggregate indicators are acceptable only if their fibres do not merge policy-incompatible states.

E7 (the seven admissible partitions). Exhaustive enumeration lists all seven admissible partitions of the fifteen — the two minimal 2-cell partitions of E3, four refinements at 3 cells, and the discrete partition — and certifies the upper-set structure (every admissible partition refines an admissible 2-cell partition).

E8 (singleton equivalence, two computations). The equivalence of Proposition 1 is verified by two independent code paths: fibre-intersection tests and one-step belief-viability of each presented fibre, instance by instance over all fifteen partitions.

What is not claimed

No sensor-cost theory is developed (the cost function is cell count); no informational-efficiency or mechanism-design claims; the delay rule is stated for the audited hold class, not for general exit-time values beyond the successor’s; the design search is exhaustive over the fifteen-partition target, not over general partition lattices (Bell-number growth is acknowledged as the combinatorial wall). Non-convexity enters only as the recorded counterinstance — no general non-convex design theory is claimed. Real monitoring systems involve noise and delay stochastics, which belong to the stochastic selector’s setting.

Declarations

Funding. None. **Competing interests.** None. **Data availability.** No external data were used. **Code availability.** The verification script `monitoring_design_v3_verify.py` is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); it reproduces all nine check families verbatim. **AI declaration.** GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review of the earlier editions; the joint audits and this revision were prepared with further AI assistance, under the programme’s verification discipline.